

PHISHING DETECTION & AWARENESS REPORT

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1. Introduction

- Phishing is a deceptive cyber attack where attackers pretend to be trusted companies, bosses, or banks.
- Their goal is to trick everyday users into handing over sensitive secrets, such as bank account details, mobile numbers, login passwords, or One-Time Passwords (OTPs).
- These scams are heavily delivered through fake emails, text messages, or malicious websites that look identical to real login screens.

2. Objectives

- To analyze live phishing email samples from the spam ecosystem.
- To identify key phishing indicators using simple visual checks.
- To perform basic email header and transit verification.
- To calculate and classify email risk levels cleanly.
- To provide easy-to-understand prevention guidelines and training rules for employees.

3. Tools Used

The following software tools and platforms were utilized during this analysis sequence:

Tool Name	Purpose
Gmail Web Client	Used to view the message and safely extract raw email text headers via 'Show Original'.
MXToolbox	Automated tool used to analyze email headers, track delivery paths, and check security rules.
MS Word / GitHub	Used for creating the final project documentation and managing the public evidence repository.

4. Email Sample 1 (Emergency Paisa)

Review Your Revised ₹50,000 Loan Approval Summary

Spam x

Emergency Paisa <notification@skill.getujobs.com>

to me

Tue, May 26, 7:0

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report not spam

INSTANT LOAN UPDATE

READY FOR DISBURSAL

₹50,000 Loan Ready

Verify your mobile number to receive the approved amount directly into your bank account.

APPROVED AMOUNT

₹50,000

INTEREST

18% p.a.

MONTHLY EMI

₹5,867

Fast Approval

Direct Transfer

Easy Process

Verify & Continue →

Header Analyzed

Email Subject: Review Your Revised ₹50,000 Loan Approval Summary

Analyze

Delivery Information

DMARC Compliant

SPF Alignment

SPF Authenticated

DKIM Alignment

DKIM Authenticated

Relay Information

Received

Delay:

0 seconds

From mail1.skill.getujobs.com to mx.google.com

to

0 0.2 0.4 0.6 0.8 1 1.2

Relay (Seconds)

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	mail1.skill.getujobs.com 103.197.16.249	mx.google.com	ESMTPS	5/26/2026 1:35:27 PM	✓
2	0 seconds		2002:a05:6022:32d1:20b0:104:486e:4634	SMTP	5/26/2026 1:35:27 PM	

Indicators Found in Sample 1:

- Display Name Spoofing:** Changing the visible sender name to mimic a trusted entity while hiding the true address.
- Sender Domain Mismatch:** A total contradiction between the company name used in the text and the actual sending domain.
- Financial Bait Lure:** Offering unexpected money or loan approvals to exploit financial desire.
- Artificial Urgency:** Creating fake deadlines to panic the user into acting without thinking.
- Generic Targeting:** Addressing the victim by their raw email address handle instead of their actual name.

- ## 5. Email Sample 2 (Bajaj Finserv)



- **Display Name Spoofing:** Changing the visible sender name to mimic a trusted financial company while hiding the true address.
- **Sender Domain Mismatch:** A total contradiction between the well-known company name used in the text and the actual job portal domain.
- **Financial Bait Lure:** Offering massive unexpected money or loan options up to ₹25,00,000 to exploit financial desire.
- **Artificial Urgency:** Using a fake 'Reminder' about a missing payment to panic the user into acting fast.

- ### 5b. Email Sample 3 (HDFC Bank Notification)



- **Brand Impersonation / Display Name Spoofing:** The sender name displays 'HDFC Bank Notification' to look authentic, but hides an entirely unrelated infrastructure domain.
- **Sender Domain Mismatch:** Severe misalignment between the targeted Indian banking entity (HDFC) and the actual sending source job portal domain.
- **Fake Transaction / Account Lure:** Traps the recipient into interacting by claiming a 'TATA HDFC Bank_Card Application_Received' update for a credit option they never applied for.
- **Generic Greeting:** Employs a non-specific 'Dear Customer' placeholder rather than utilizing the formal account owner name.
- **Obfuscated Call to Action:** Forces user interaction via a high-urgency 'Confirm Card Details to Continue' link routing through external tracking setups.

- **Shared Backend Fingerprinting:** Utilizes the exact same server cluster IP and tracking campaign logic as the previous malicious samples, confirming an organized threat campaign.

6. How the Attack Works

- 1. The Setup:** The attacker configures a high-volume automated distribution layout using an established domain infrastructure that they control or have fully compromised.
- 2. The Bait:** They construct enticing, template-driven messages leveraging severe financial rewards (e.g., instant loan approvals or pre-calculated credit disbursements) engineered to evoke panic or high excitement.
- 3. The Trap:** Prominent call-to-action nodes (e.g., 'Verify Now' or 'View Loan Details' hyperlinks) route the victim away from safe email boundaries onto external tracking infrastructure masks.
- 4. The Theft:** When the victim submits input details—such as mobile phone verifications, identity indexes, or session credentials—the parameters are parsed directly into server databases owned exclusively by the fraud syndicate.

7. Risk Scoring & Classification System

Indicator Type	Sample 1	Sample 2	Sample 3	Risk Points
Suspicious Sender Domain	Yes	Yes	Yes	2
Urgent Language or Deadlines	Yes	Yes	Yes	1
Hidden or Mismatched Hyperlinks	Yes	Yes	Yes	2
Unexpected Financial Lure	Yes	Yes	Yes	2
Generic Greeting / Email Handle	Yes	Yes	Yes	1
Domain Security Failure (DMARC)	Yes	Yes	Yes	2
Total Risk Evaluation Score	10 / 10	10 / 10	10 / 10	High Risk

SPF and DKIM Information

dmarc:skill.getujobs.com

Hide

Solve Email Delivery Problems

v=DMARC1; p=none; sp=none; adkim=r; aspf=r; ruf=mailto:dmarc@mlrcloud.com; rua=mailto:dmarc@mlrcloud.com

Risk Level Matrix Key:

- 0 – 3 Points: Low Risk (Safe / Regular Spam Category)
- 4 – 6 Points: Medium Risk (Suspicious / Marketing Traffic)
- 7 – 10 Points: High Risk (Phishing / Dangerous Fraud Vector)

8. Prevention Guidelines (User Safety Tips)

- **Check the True Address:** Never trust just the name at the top of an email. Click or tap on the sender's name to see the actual email address behind it.
- **Hover Before You Click:** If you are on a computer, hover your mouse arrow over any button or link without clicking it. Look at the bottom corner of your screen to see the true website address it wants to take you to.
- **Go There Yourself:** If an email says your account has an issue, do not use the links inside the email. Open a fresh internet browser window, type the official website address yourself, and log in securely.

9. Employee Do's and Don'ts

DO:

- Report any email that forces a strict time limit or feels unusual to your internal security team.
- Check if a financial email addresses you by your actual full legal name instead of your email user handle or generic placeholders.
- Delete unsolicited messages offering unexpected money, credit options, or card approvals immediately.

DON'T:

- Don't click 'Unsubscribe' inside an obvious phishing email, as this just tells the attacker your mailbox is active and ready for more spam.
- Don't assume an email is safe just because it has an official company logo or look-alike branding.
- Don't ever type your password, phone number, or OTP into a web page you reached through an email link.

10. Conclusion

- Phishing scams remain a highly dangerous tool used to breach user privacy and steal digital funds.
- By performing basic header verifications and learning how to match sender domains against body text, users can instantly neutralize incoming threats.
- Regular security awareness, simple checks, and reporting suspicious activity are the best defenses to keep digital workplaces safe.

GITHUB REPOSITORY LINK: https://github.com/adi783885/FUTURE_CS_02